

MACHINE LEARNING PROJECT

PREDICTING ROAD TRAVEL TIME USING RIDGE AND LASSO REGRESSION

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SECTION-8

BATCH-11

PROBLEM STATEMENT

Travel time for a road journey depends on a mix of numerical and categorical factors — distance, average speed, number of traffic signals, traffic congestion level, weather conditions, road type, vehicle type, day of the week, and time of day. Estimating travel time accurately is essential for route planning, traffic management, and logistics scheduling.

Since several of these features are related to each other — for example, traffic level, number of signals, and average speed often move together — a plain linear regression model can suffer from multicollinearity, leading to unstable coefficients and poor generalization to new data. This project aims to build a regression model that predicts travel time (in minutes) using the available trip features, and applies Ridge and Lasso regularization to handle correlated features, improve model stability, and identify which features are most influential in determining travel time.

01

Multiple factors affect travel time — distance, speed, signals, traffic, weather, road type, vehicle type, day, and time all play a role.

02

Correlated features cause instability — traffic level, signals, and speed often move together, making plain linear regression unreliable.

03

Ridge and Lasso fix this — they stabilize the model and reveal which features matter most for predicting travel time.

WHY TRAVEL TIME PREDICTION MATTERS

Used across real-world transportation and mobility platforms everyday



01.

Helps drivers and navigation apps (Google Maps, GPS systems) suggest the fastest route and give accurate arrival times.



02.

Helps city authorities monitor congestion, plan signal timings, and manage road infrastructure.



03.

Companies use it to plan delivery routes, driver allocation, and estimate accurate delivery windows.



04.

Better time estimates help reduce unnecessary idling, fuel consumption, and operational costs for fleets.

DATASET

My dataset contains trip-level records with numerical features like distance, average speed, and number of signals, along with categorical features like traffic level, weather, road type, vehicle type, day of week, and time of day. The target variable is `Travel_Time_Minutes`

Target Variable:

`Travel_Time_Minutes` — the value being predicted

Numerical Features:

- **`Distance_km`** — distance of the trip in kilometers
- **`Avg_Speed_kmph`** — average speed during the trip
- **`Number_of_Signals`** — number of traffic signals encountered

Categorical Features:

- **`Traffic_Level`** — level of traffic congestion (e.g., Low/Medium/High)
- **`Weather`** — weather condition during the trip
- **`Road_Type`** — type of road traveled on
- **`Vehicle_Type`** — type of vehicle used
- **`Day_of_Week`** — day the trip occurred
- **`Time_of_Day`** — time period of the trip (e.g., Morning/Afternoon/Evening)

FEATURE ENGINEERING

Feature engineering prepares raw trip data for modeling by converting categorical variables into numeric form and standardizing numerical features. This step also involves creating additional derived features and checking for multicollinearity, ensuring the dataset is suitable for Ridge and Lasso regression.

01.

One-Hot Encoding applied to Traffic_Level, Weather, Road_Type, Vehicle_Type, Day_of_Week, and Time_of_Day — converts each category into binary (0/1) columns since Ridge/Lasso require numeric input.

02.

StandardScaler applied to numerical features — Distance_km, Avg_Speed_kmph, Number_of_Signals — since Ridge/Lasso penalize coefficient size, and unscaled features would be penalized unevenly.

03.

New features created to strengthen prediction — e.g., Signals per km (signal density along the route) and Is_Peak_Hour (binary flag for rush-hour trips based on Time_of_Day).

04.

Correlation heatmap used to examine relationships between Distance_km, Avg_Speed_kmph, and Number_of_Signals — confirming why regularization (Ridge/Lasso) is needed for this dataset.

METHODOLOGY

The methodology follows a structured pipeline — starting with data cleaning and exploratory analysis, followed by feature engineering to prepare the data for modeling. Linear, Ridge, and Lasso regression models are then trained and tuned using cross-validation, and finally compared using standard evaluation metrics to identify the best-performing and most interpretable model.

01

Checked missing values, duplicates, and data types; explored travel time patterns using histograms, boxplots, scatter plots, and a correlation heatmap.

02

Applied One-Hot Encoding to categorical features (traffic level, weather, road type, etc.) and StandardScaler to numerical features (distance, speed, signals).

03

Split data into train/test sets; trained Linear Regression (baseline), Ridge, and Lasso models; tuned regularization strength (alpha) using cross-validation.

04

Compared models using RMSE, MAE, and R^2 score; analyzed coefficient shrinkage to identify the most influential features affecting travel time.

RIDGE VS LASSO

Aspect	Ridge Regression (L2)	Lasso Regression (L1)
Penalty	Sum of squared coefficients ($\lambda \sum \beta^2$)	Sum of absolute coefficients ($\lambda \sum$
Effect on coefficients	Shrinks all coefficients smoothly toward zero	Can shrink some coefficients to exactly zero
Feature selection	No — keeps all features in the model	Yes — automatically removes less useful features
Best used when	Most features are useful but correlated	Some features are irrelevant/redundant
Interpretability	Less interpretable (all features retained)	More interpretable (sparse model)
Stability with correlated features	More stable — distributes weight across correlated features	Can arbitrarily pick one feature among correlated ones
Geometric shape of constraint	Circular (smooth)	Diamond-shaped (sharp corners on axes)

CONCLUSION

This project successfully developed a machine learning regression model to predict travel time using trip-related features such as distance, average speed, number of traffic signals, traffic level, weather, road type, vehicle type, day of week, and time of day. Since several of these features were correlated with one another, plain linear regression showed instability, motivating the use of Ridge and Lasso regularization techniques.

Both Ridge and Lasso regression improved model stability and generalization compared to the baseline linear regression model, effectively addressing the issue of multicollinearity. Lasso regression additionally provided valuable interpretability by shrinking less important feature coefficients to zero, helping identify the features that most strongly influence travel time — such as distance, traffic level, and number of signals.

Overall, this project demonstrates that regularized linear regression techniques are a practical and interpretable approach for travel time prediction, offering a strong foundation that can be extended in future work using more advanced models such as Random Forest or XGBoost, or deployed as a real-time prediction service.

**THANK YOU
VERY MUCH**